PACDIS framework

MOEA loop

Latin hypercube sampling

Objective evaluation
Initialize archive and FE

$FE < FE_{max}$?

Yes

No

Return archive

Current population $\mathcal{P}$

Objective evaluation
Update archive and FE

Environmental selection

PAQC

Evaluated population $\mathcal{P}$

A B C D

Labels from representatives

L_i

A	1
B	0
C	1
D	0

Scores from directions of nondominated solutions

S_i

A	0.60
B	0.90
C	0.40
D	0.20

Scores

A	0.7
B	0.675
C	0.55
D	0.15

S_i term L_i term

Rank: A → B → C → D

$\mathcal{C}_1$: A

$\mathcal{C}_2$: B C D

Train indicator model
RBF-SVR $\mathcal{M}_I$
on population $\mathcal{P}$

Model training

C1	C1	0
C1	C2	+1
C2	C1	-1
C2	C2	0

Relation model $\mathcal{M}_R$

CDIS

Choose one mode m

Relation-guided search
and candidate screening

$m = \text{ind}$

Indicator prediction

A	0.82
B	0.63
C	0.91

Rank: C → A → B

$m = \text{amb}$

Ranking with ambiguity

A	1.00
B	1.10
C	0.12

Relation score $\bar{R}$ Ambiguity term $0.3\bar{U}$

Rank: B → A → C

Illustrative scores

Promising solutions